

“Beyond the Label: $\delta^{15}\text{N}$ Confirms Farming System While $\delta^{13}\text{C}$ Reflects Environmental Backgrounds in Tomatoes”

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ABSTRACT

The carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopic compositions of tomato products were investigated to assess their potential to distinguish between farming systems and geographic origins. Using isotope ratio mass spectrometry and applying statistical methods (Welch's t-tests, ANOVA, PCA, and LDA), revealed that organic tomatoes showed higher $\delta^{15}\text{N}$ values due to manure fertilization, while lower $\delta^{15}\text{N}$ values in conventional tomatoes are indicative of using synthetic fertilizers. However, regional baselines limited effective discrimination of farming types. $\delta^{13}\text{C}$ values were significantly influenced by environmental factors and farming interactions, while combining $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ improved classification accuracy of production methods; nevertheless, additional isotopic systems are needed for precise geographic origin identification.

1. Introduction

The growing demand for organic produce has heightened the need for robust analytical methods to verify labeling and trace the origin of products. Stable isotope ratio analysis has emerged as a powerful approach in food authentication, since the isotopic composition of plants reflects both nutrient inputs and environmental conditions (Kelly et al. 2005). Among these, carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopes are particularly relevant to agricultural systems and are increasingly applied in food authenticity studies (Bontempo et al. 2020).

Nitrogen isotopes offer insight into the sources of fertilizers. Synthetic nitrogen fertilizers produced via the Haber–Bosch process generally have $\delta^{15}\text{N}$ values near 0‰, while organic amendments, such as manures and composts, have higher $\delta^{15}\text{N}$ values than synthetic fertilizers (due to microbial processing and volatilization) (Bateman & Kelly 2007). Consequently, crops grown with organic fertilizers often show higher $\delta^{15}\text{N}$ values than those cultivated with synthetic inputs. Tomato studies confirm this pattern; (Nakano et al. 2003) observed a $\delta^{15}\text{N}$ value of +7.1‰ under organic fertigation versus +0.3‰ with inorganic fertilizer, and (Trandel et al. 2018) reported consistently higher $\delta^{15}\text{N}$ values in greenhouse tomatoes amended with organic matter. However, our findings align with this pattern, showing a highly significant enrichment of +5–6‰ in organic tomato products relative to conventional ones, with almost no overlap between groups. These findings demonstrate $\delta^{15}\text{N}$ as a robust

discriminator of production method. In contrast, $\delta^{13}\text{C}$ is less effective for distinguishing between farming systems, as it primarily reflects plant water-use efficiency and environmental stress, rather than the type of fertilizer used (Nakano et al. 2003; Bontempo et al. 2020). We also found similar results in our study; value differences in $\delta^{13}\text{C}$ by farming type were marginal and only regionally consistent (e.g., Bavarian samples), highlighting environmental drivers as the primary source of variability.

Beyond production methods, stable isotopes also carry regional signals. Climate, soil type, and hydrology can alter baseline isotopic compositions ($\delta^{15}\text{N}$, $\delta^{13}\text{C}$), independently of farming system (Amundson et al. 2003). Multi-isotope approaches have been successful in distinguishing the regional origins of foods. For instance, (Bontempo et al. 2011) achieved high accuracy in classifying Italian tomato products using $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, $\delta^{34}\text{S}$, and mineral profiles, while (Kelly et al. 2005) highlighted the value of isotopes in geographic traceability across diverse commodities. However, in our dataset, bulk $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ achieved only ~65% accuracy for regional classification, underscoring their limited power for origin tracing when used alone. $\Delta^{13}\text{C}$ value clearly distinguished more arid from humid regions, while $\delta^{15}\text{N}$ value varied with both fertilizer type and soil baselines, resulting in partial confounding of farming and regional effects, possibly due to imbalanced sampling. Moreover, as (Bontempo et al. 2020) showed for tomato passata, bulk $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values alone provide intense discrimination of farming methods but limited power for clear regional separation, underscoring the importance of compound-specific or multi-isotope approaches.

We therefore address two overarching questions: (1) Can $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values reliably differentiate tomato products by farming type across different regions? (2) To what extent is the isotopic signature driven by production method versus geographic/environmental factors? By sampling tomato fruits and derived products from multiple regions and cultivation systems, this study evaluates isotopic ratios under real-world conditions and discusses the potential and limitations of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values as markers of organic provenance.

2. Materials and Methods

2.1 Sample collection and preparation:

In this study, tomato products (A–J) representing both organic and conventional farming systems were purchased from major European retailers (Aldi, Kaufland, Ebl Naturkost, and EDEKA; **Table 1**). Samples originated from three central production regions: Bavaria (Germany), Almería (Spain), and Souss-Massa (Morocco). Each product was subsampled in quadruplicate (except for B($n = 12$), D($n = 3$), and J($n = 9$), resulting in a total of $n = 52$ samples. All samples were freeze-dried, homogenized with a mortar and pestle, and stored airtight. Approximately $1000 \mu\text{g} \pm 100 \mu\text{g}$ of homogenized powder was weighed into tin capsules (IVA, Germany) for analysis.

2.2 Isotopic and elemental analysis:

Stable isotope ratios of carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$), together with elemental composition (%C, %N, and C:N ratio), were analyzed using a Flash 2000 Elemental Analyzer (Thermo Scientific) coupled via a Conflo II interface to a Delta V Advantage IRMS (Thermo Scientific). Isotope values are expressed in δ notation (‰) relative to Vienna Pee Dee Belemnite (VPDB) for carbon and atmospheric N_2 (AIR) for nitrogen. Calibration was

performed against international reference standards (IAEA-N1, IAEA-N2, USGS-41, IAEA-CH7) and a laboratory standard ("Peptone II"), with elemental quantification calibrated using atropine and cyclohexanone-2,4-dinitrophenylhydrazone.

2.3 Statistical analysis:

All analyses were performed in R (v4.4.2). Farming system comparisons were evaluated with Welch's t-tests (reporting t, df, and p). Regional effects were tested using one-way ANOVA, while combined two-way ANOVA models assessed the influence of Region, Farming Type, and their interaction. For ANOVAs, F-statistics, p-values, and η^2 effect sizes were extracted. Multivariate structure was explored using principal component analysis (PCA) on z-standardized isotopic and elemental variables ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$, %C, %N, C:N), and classification performance was assessed using linear discriminant analysis (LDA). PCA variance explained by the first components and LDA classification accuracies for farming type and region were reported in the results section. Statistical significance was set at $\alpha = 0.05$.

3. Results

3.1 Comparative Isotopic and Elemental Profiles:

Isotopic differences were observed between organic and conventional tomato products, whereas basic elemental composition broadly overlaps. For $\delta^{13}\text{C}$ (**Figure 1**, top left), $\delta^{13}\text{C}$ was slightly higher in organic tomatoes (mean $-29.2 \pm 1.0\text{‰}$) than in conventional ones ($-30.5 \pm 1.1\text{‰}$), indicating $\delta^{13}\text{C}$ values were $\sim 1.4\text{‰}$ higher in organic management. Along with partial overlap between groups, the trend was found as ($t = 2.10$, $p = 0.070$).

In contrast, (**Figure 1**, top right), $\delta^{15}\text{N}$ showed a pronounced difference: organic tomatoes averaged around $+5.9 \pm 2.0\text{‰}$, while conventional tomatoes were near $+0.1 \pm 0.6\text{‰}$, a gap of $\sim 5\text{--}6\text{‰}$ that was highly significant ($t = 6.12$, $p = 0.002$). There was essentially no overlap in $\delta^{15}\text{N}$ between the two groups.

Elemental traits (%N and C:N ratio; **Figure 1**, bottom panels) did not differ significantly between farming systems. Both groups, Organic and Conventional products, had similar %N ($1.7 \pm 0.2\%$ vs. $1.7 \pm 0.3\%$, $p = 0.761$) and overlapping C:N ratios (24.4 ± 3.1 vs. 28.3 ± 8.35 , $p = 0.367$).

3.2 $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ across Farming Types and Regions:

To disentangle farming effects from geographic variation, we examined $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ at the regional level (**Table 2**, **Figure 2**). In Bavaria, where we had both farming types, Organic samples averaged $\delta^{13}\text{C} -29.3\text{‰}$ and $\delta^{15}\text{N} +6.3\text{‰}$, compared to $\delta^{13}\text{C} -30.8\text{‰}$ and $\delta^{15}\text{N} +0.2\text{‰}$ for Conventional samples. The Bavarian organic tomato products had a higher $\delta^{15}\text{N}$ of $\sim 6\text{‰}$ relative to their conventional counterparts.

In Almería, only an organic product was available ($n = 12$); it showed $\delta^{13}\text{C} -28.7\text{‰}$ and $\delta^{15}\text{N} +4.5\text{‰}$. This Almería organic sample had a higher $\delta^{13}\text{C}$ (less negative) than any Bavarian sample. Its $\delta^{15}\text{N}$ value of $+4.5\text{‰}$ was slightly lower than the Bavarian organic mean, but still well above any conventional value. In Souss-Massa, only a conventional product ($n = 9$) was available, which showed $\delta^{13}\text{C} -29.6\text{‰}$ and $\delta^{15}\text{N} -0.2\text{‰}$. This $\delta^{15}\text{N}$ value is near zero,

as expected for synthetic fertilizer usage, and its $\delta^{13}\text{C}$ (-29.6‰) is higher (less negative) than the Bavarian conventional (-30.8‰).

Figure 2(A) visualizes these findings more clearly: $\delta^{15}\text{N}$ values cluster tightly by farming type across regions, with all organic samples ($\sim +4$ to $+6\text{‰}$) and all conventional samples near 0‰ . By contrast, $\delta^{13}\text{C}$ values overlap substantially between farming systems. **Figure 2(B)**, ($\delta^{13}\text{C}$ vs. $\delta^{15}\text{N}$ scatter) makes this even clearer: all organic products lie well above the conventional ones along the $\delta^{15}\text{N}$ axis, while the $\delta^{13}\text{C}$ axis exhibits regional variation (Almería's organic samples have higher $\delta^{13}\text{C}$ than Bavaria's, and the Souss-Massa conventional sample has slightly higher $\delta^{13}\text{C}$ than the Bavarian conventional).

3.3 Effects of Region and Farming Type (ANOVA):

We used factorial ANOVA to formally assess the influence of Region, Farming Type, and their interaction on tomato's $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values (**Table 3**). The two-way ANOVA results revealed a significant effect of Region on both isotopes, with differing interaction patterns. For $\delta^{13}\text{C}$, Region had a highly significant impact ($F = 39.93$, $p < 0.001$), and there was also a strong Region $\times$ Farming Type interaction ($F = 28.15$, $p < 0.001$). In contrast, the Farming Type main effect on $\delta^{13}\text{C}$ was not significant (no estimate). For $\delta^{15}\text{N}$, Region was the only significant factor ($F = 8.18$, $p = 0.029$). Neither Farming type ($F = 0.09$, $p = 0.764$) nor the Region $\times$ Farming Type interaction ($F = 0.08$, $p = 0.925$) reached significance.

To further clarify, we performed supporting one-way ANOVAs (**Table 4**) for Farming Type (ignoring Region) and for Region (ignoring Farming Type). The one-way results confirmed what the two-way model obscured. Across the total dataset, Farming Type had a highly significant effect on $\delta^{15}\text{N}$ ($F = 38.49$, $p < 0.001$), explaining roughly 43.5% of the total variance. By contrast, farming type's effect on $\delta^{13}\text{C}$ was much weaker ($F = 3.54$, $p = 0.065$). Conversely, Region alone was a significant determinant of both $\delta^{13}\text{C}$ ($F = 8.25$, $p < 0.001$, $\sim 25\%$ variance explained) and $\delta^{15}\text{N}$ ($F = 11.35$, $p < 0.001$, $\sim 32\%$ variance explained).

3.4 Multivariate Separation of Isotopic and Elemental Traits:

Principal component analysis (PCA) was used to explore joint variation in $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, %N, %C, and C:N (**Figure 3**). The first two axes explained 73.2% of the variance (PC1 = 43.9%, PC2 = 29.3%). PC1 mainly reflected elemental composition, with positive loadings for C:N (+0.60) and %C (+0.48) and negative loadings for %N (-0.43) and $\delta^{15}\text{N}$ (-0.45). This "compositional axis" distinguished samples by nutrient profile and product type rather than farming regime; for example, the Souss-Massa conventional tomato scored high on PC1 due to its dried form and elevated C:N. PC2, by contrast, served as the "isotopic axis," with $\delta^{13}\text{C}$ (+0.63) and $\delta^{15}\text{N}$ (+0.45) loading positively. It clearly separated Organic from Conventional samples, where Organic tomatoes (high in $\delta^{15}\text{N}$ and relatively low in $\delta^{13}\text{C}$ values) scored high, while Conventional samples scored low (**Figure 3**).

Finally, we applied linear discriminant analysis (LDA) using $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ to classify samples by farming type. The LDA achieved 95–96% correct classification of farming method; 23 of 24 conventional samples and 27 of 28 organic samples were correctly identified (**Table 5**). In contrast, regional classification was weak (65.4% accuracy). Samples from Bavaria and Almería were often misclassified by origin.

4. Discussion

4.1 *Isotopic and Elemental Structure:*

The summary of isotopic and elemental traits suggests that organically grown plants had marginally higher water-use efficiency, likely due to reduced irrigation or water stress, resulting in higher $\delta^{13}\text{C}$ values in plant tissue. This can eventually lead to stomatal closure and high $\delta^{13}\text{C}$ (Bontempo et al. 2020). Moreover, organic farming often aims to conserve soil moisture and water resources (especially under drought conditions), and lower nitrogen availability in organic fields may also induce stronger water-use efficiency and higher $\delta^{13}\text{C}$ (Paolini et al. 2015).

The higher $\delta^{15}\text{N}$ values in organic products reflect the use of organic fertilizers (manure or compost), which leaves crops with higher ^{15}N levels, compared to synthetic N fertilizers in conventional farming, which have $\delta^{15}\text{N}$ values close to 0‰ (Bateman & Kelly 2007). These results are consistent with previous studies showing $\delta^{15}\text{N}$ as a reliable indicator of fertilizer source and organic management (Nakano et al. 2003; Bateman et al. 2005); (Trandel 2019).

Our results show that such elemental measures of nutrient content lack discriminatory power relative to isotope markers: a tomato's %N alone does not reveal how that N was sourced. This suggests that the overall nutrient status of the tomatoes was similar across both farming methods, likely because both cultivation systems provided sufficient nitrogen to the plants. The comparable %N and C:N underscore that an organic tomato can be just as nutrient-rich as a conventional one; what differs is the source and isotopic signature of that nitrogen.

4.2 *$\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ vs Methods and Origin:*

To explore the interaction between $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ in relation to farming type and region, our findings reveal distinct $\delta^{15}\text{N}$ differences within a single region, reinforcing the isotopic imprint of farming practices as a testament to the substantial impact of manure-based versus synthetic fertilizers on plant $\delta^{15}\text{N}$. We also found that the same farming system within the different regions, for example, both Bavarian and Almerian organic products, showed slight differences in $\delta^{15}\text{N}$ values. This finding also suggests that there are significant differences in nitrogen sources, reflected in $\delta^{15}\text{N}$ values. Likewise, (Choi et al. 2017) note that animal manures and composts “have higher $\delta^{15}\text{N}$ values” because organic processes remove ^{14}N , leaving the remaining material relatively ^{15}N -rich. Therefore, we can infer that changes in $\delta^{15}\text{N}$ values within the same farming type may occur due to different sources of ^{15}N .

The $\delta^{13}\text{C}$ difference also suggests that the organic Bavarian crops experienced drier or more water-limited conditions, which could reflect different irrigation strategies or microclimates between the farming systems. Our findings are consistent with previous research. (Kirkels et al. 2022) observed that $\delta^{13}\text{C}$ values of C_3 vegetation in the Godavari basin (India) became significantly higher under drier conditions, with mean annual precipitation emerging as a dominant control on plant $\delta^{13}\text{C}$. Similarly, (Fischer & Du Toit 2019) reported that higher (less negative) $\delta^{13}\text{C}$ values in tree rings were associated with reduced water availability, reflecting higher water-use efficiency under drought stress in plantation forests.

4.3 ANOVA Test:

The two-way ANOVA outcome indicates that the $\delta^{13}\text{C}$ value response to organic vs. conventional farming varied by region, meaning that farming type per se did not shift $\delta^{13}\text{C}$ values uniformly across all sites, which makes sense given our earlier observation (**Section 3.2**) that the organic–conventional $\delta^{13}\text{C}$ value difference was evident in Bavaria but confounded by the regions.

By contrast, the two-way ANOVA revealed that region was the only significant factor for $\delta^{15}\text{N}$. This initially seems surprising, given the distinct $\delta^{15}\text{N}$ values observed between organic and conventional samples, but it reflects our unbalanced sampling design; Organic samples came from two particular regions (Bavaria and Almería), and Conventional samples from Bavaria and Souss-Massa, so the Farming Type and Region effects were partially confounded (for example, there was no Conventional sample from Almería). In such a case, the ANOVA cannot isolate a pure farming effect (hence “no estimate” for Farming Type), though this does not mean farming type had no effect on $\delta^{15}\text{N}$, only that it could not be estimated in the two-way model.

Our one-way ANOVAs clarify this issue. Across the entire dataset, Farming Type had a highly significant effect on $\delta^{15}\text{N}$ values ($F = 38.49$, $p < 0.001$) and a weaker effect on $\delta^{13}\text{C}$, consistent with the earlier t-test results. The region alone was significant for both isotopes, explaining approximately 25% of the $\delta^{13}\text{C}$ variance and 32% of the $\delta^{15}\text{N}$ variance. Thus, these one-way analyses reveal why the two-way ANOVA yielded unexpected results. In other words, if one ignores the region (or vice versa), the $\delta^{15}\text{N}$ distinction between farming systems is highly significant, matching our univariate findings and many previous studies (Bateman & Kelly 2007; Trandel et al. 2018). Meanwhile, the $\delta^{13}\text{C}$ value results confirm that carbon isotopes contributed less to distinguishing farming types and more to distinguishing regions, likely because climatic factors (such as rainfall and humidity) primarily drive $\delta^{13}\text{C}$ value differences, as found by (Bontempo et al. 2020).

4.4 PCA and LDA Test:

These multivariate results clearly reinforce our univariate findings. The PCA indicates that isotopic markers, especially $\delta^{15}\text{N}$, encode farming approach, whereas elemental traits capture product or regional differences. In particular, the inverse %N– $\delta^{15}\text{N}$ pattern reflects underlying fertilizer dynamics; synthetic fertilizers are rich in readily available (^{14}N) nitrogen, which boosts %N but dilutes $\delta^{15}\text{N}$ values, whereas organic amendments supply less available N that is a high ^{15}N value (Trandel et al., 2018; Bateman et al., 2005).

The LDA results further emphasize the strength of these markers. The proximate classification of farming type emphasizes $\delta^{15}\text{N}$ (and $\delta^{13}\text{C}$) as powerful authenticity indicators. The few misclassifications highlight cases where conventional farms may use manure or organic farms use legumes, demonstrating the value of multi-isotope approaches (Bontempo et al. 2020). In contrast, the inadequate regional classification confirms that $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ by themselves do not provide unique geographic signatures; samples from Almería and Bavaria were frequently confused. Thus, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are excellent predictors of production method but not of origin, unless supplemented by additional isotopic systems (Choi et al. 2017b).

5. Conclusion

The isotopic data reveal that the $\delta^{15}\text{N}$ value serves as a distinct marker of production method, while the differences in $\delta^{13}\text{C}$ values between farming systems are minimal. Notably, tomatoes grown using organic methods consistently showed higher $\delta^{15}\text{N}$ values compared to their conventional counterparts, whereas $\delta^{13}\text{C}$ alone did not effectively distinguish between cultivation types. The one-way ANOVA analysis confirmed that $\delta^{15}\text{N}$ is a strong discriminator of farming practices, while the variations in $\delta^{13}\text{C}$ values were found to be marginal. Additionally, geographic region emerged as the primary source of variance for $\delta^{13}\text{C}$ and contributed some variance to $\delta^{15}\text{N}$ values, highlighting the influence of environmental and climatic factors. Two-way ANOVA also addressed our research questions by indicating a significant interaction between region and farming for $\delta^{13}\text{C}$, in contrast to $\delta^{15}\text{N}$, which was less effective in differentiating farming types and the interaction between region and farming type due to the imbalanced sample array. Overall, $\delta^{15}\text{N}$ values proved to be the most reliable indicator of organic versus conventional farming in terms of isotopes. Nevertheless, future research should integrate these isotopes into multivariate models alongside additional tracers, ensuring an adequate sample that captures geographic and climatic signals to enhance origin traceability.

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Figures and Tables

Table 1. Sample inventory: Tomato products grouped by producer with replicate codes.

Sample Code	Supplier	Farming System	Region of Origin	Replicates (n)
A	Aldi	Organic	Bavaria (Germany)	4
B	Aldi	Organic	Almería (Spain)	12
C	Aldi	Conventional	Bavaria (Germany)	4
D	Kaufland	Conventional	Bavaria (Germany)	3
E	Kaufland	Conventional	Bavaria (Germany)	4
F	Kaufland	Conventional	Bavaria (Germany)	4
G	Ebl Naturkost	Organic	Bavaria (Germany)	4
H	Ebl Naturkost	Organic	Bavaria (Germany)	4
I	Ebl Naturkost	Organic	Bavaria (Germany)	4
J	EDEKA	Conventional	Souss-Massa (Morocco)	9
Total		—	—	52

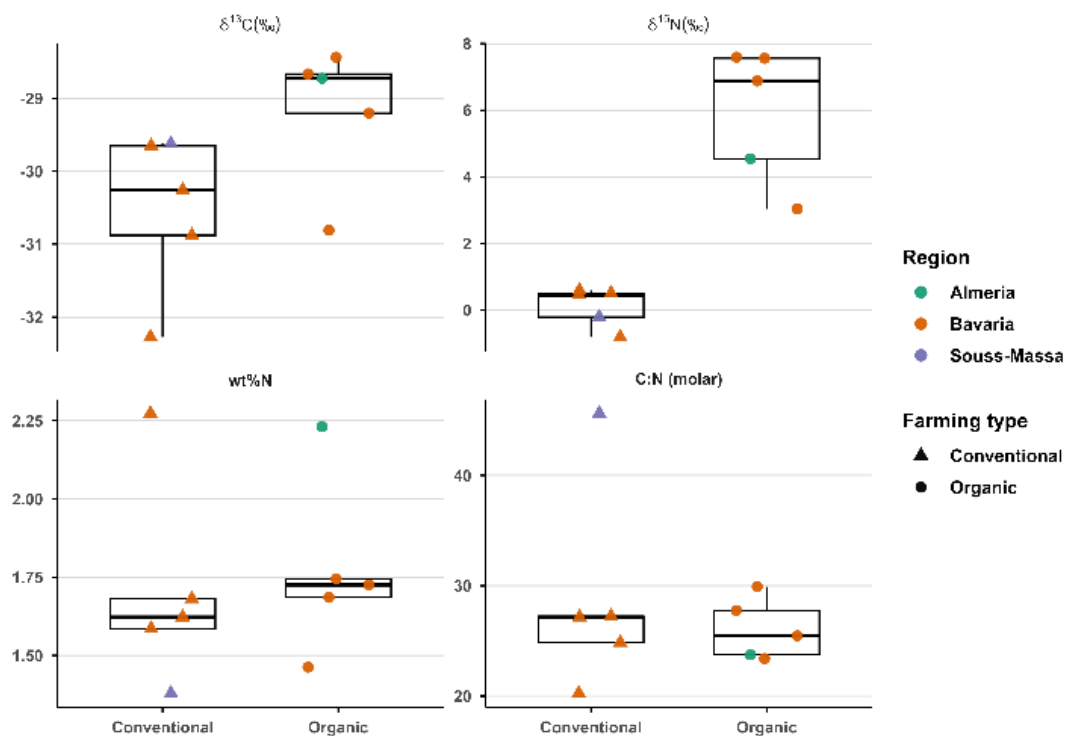


Figure 1: Isotopic ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$) and elemental (%N, C:N) traits in product group and farming type.

Table 2: Product type and region level $\delta^{13}\text{C}$, $\delta^{15}\text{N}$ with mean, $\pm\text{SD}$

region	farming type	n	d13C_mean	d13C_sd	d15N_mean	d15N_sd
Almeria	Organic	12	-28.724	0.837	4.542	0.438
Bavaria	Conventional	15	-30.840	1.475	0.157	0.809
Bavaria	Organic	16	-29.279	1.049	6.266	2.051
Souss-Massa	Conventional	9	-29.623	0.384	-0.211	0.848

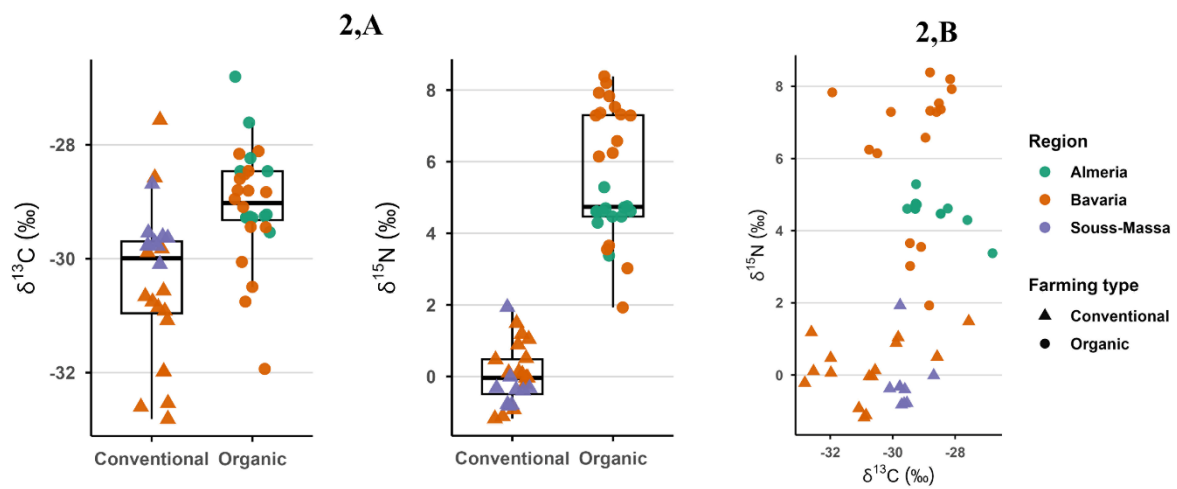


Figure 2: (A) $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ distributions by farming type, (B) Scatterplot of $\delta^{13}\text{C}$ vs $\delta^{15}\text{N}$ by region and farming type (replicates; regions in color, farming type by shape).

Table 3: Two-way ANOVA for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Region, Farming type, and interaction)

$\delta^{13}\text{C}$				
Source of variation	df	Sum of squares	F	p-value
Region	2	133.308	39.927	<0.001 ***
Farming type	1	(no est.)	—	—
Region $\times$ Farming type	2	93.974	28.146	<0.001 ***
Residual	48	80.131	—	—
$\delta^{15}\text{N}$				
Source of variation	df	Sum of squares	F	p-value
Region	2	96.02	8.18	0.029 *
Farming type	1	0.55	0.09	>0.1
Region $\times$ Farming type	2	0.89	0.08	>0.1
Residual	48	281.99	—	—

(n.s. = not significant, $p < 0.05$ = *significant)

Table 4: One-way ANOVA for $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ by farming type and Region (replicates).

Source	df	Sum of squares	F	p-value
$\delta^{13}\text{C}$				
Farming type	1	14.11	3.54	0.065 ns
Residual	50	199.42	—	—
$\delta^{15}\text{N}$				
Farming type	1	305.74	38.49	<0.001 ***
Residual	50	397.5	—	—

Source	df	Sum of squares	F	p-value
$\delta^{13}\text{C}$				
Region	2	124.27	8.25	<0.001 ***
Residual	49	368.98	—	—
$\delta^{15}\text{N}$				
Region	2	408.65	11.35	<0.001 ***
Residual	49	882.16	—	—

(n.s. = not significant, $p < 0.05$ = *significant)

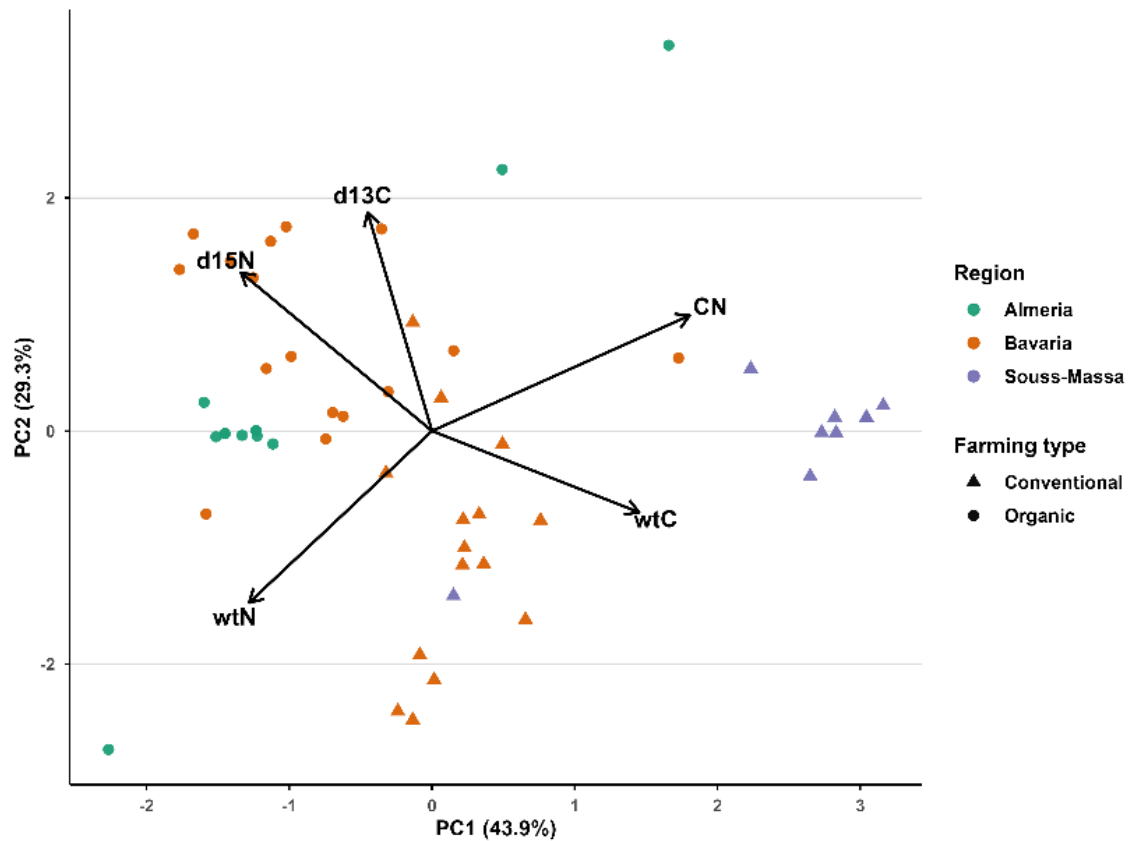


Figure 3: PCA biplot of isotopic and elemental traits ($\delta^{13}\text{C}$, $\delta^{15}\text{N}$, %N, %C, C:N), showing variable loadings and sample clustering by region and farming type.

Table 5: LDA classification of farming type (Organic vs Conventional) and region (Bavaria, Almería, Souss-Massa).

True class	Predicted Conventional	Predicted Organic	Row accuracy (%)
Conventional	23	1	95.8
Organic	1	27	96.4
Overall accuracy = 96.2% (50/52 correct classifications)			

True region	Pred. Almería	Pred. Bavaria	Pred. Souss-Massa	Row accuracy (%)
Almería	2	10	0	16.7
Bavaria	2	27	2	87.1
Souss-Massa	0	4	5	55.6
Overall accuracy = 65.4%				